End-to-End E-Commerce Analytics & Churn Prediction Pipeline

Dataset: Olist Brazilian E-Commerce Public Dataset

Domain: Marketplace Analytics & Customer Retention

Tech Stack: Python (Pandas, Scikit-Learn, SciPy, Matplotlib, Seaborn), SQLite, DBeaver

1. Project Overview

This project implements an end-to-end data engineering and predictive analytics pipeline using the Olist e-commerce marketplace dataset. Because online marketplaces are heavily characterized by high one-time buyer rates (churn), this analysis focuses on defining, engineering, predicting, and statistically validating the key drivers of customer retention.

2. Architecture & Data Pipeline (Medallion Pattern)

The project follows a modern data engineering medallion architecture implemented locally via SQLite and Python:

- **Bronze Layer (Ingestion):** Raw CSV tables (customers, orders, order_items, products, sellers, payments, reviews, and translations) loaded directly into SQLite (ecommerce.db).
- **Silver Layer (Cleaning & Transformation):** Pandas scripts used to handle missing values, sanitize strings, convert raw timestamps to proper datetimes, and compute row-level metrics like delivery_delay_days and total_item_cost.
- **Gold Layer (Feature Store):** Aggregated customer-level behavioral and operational profiles (gold_customer_churn_features) built to fuel machine learning and statistical evaluations.

3. Key Churn Findings & Insights

- **The Scale of Churn:** Over 97% of Olist customers are one-time buyers, making understanding retention friction critical.
- **Logistics Impact:** Supply chain failures—specifically missed delivery windows (delivery_delay_days > 0)—severely degrade customer lifetime value and drive permanent churn.
- **Review Sentiment:** Low average review scores (review_score) strongly correlate with a lack of repeat purchases.

4. Predictive Modeling (Machine Learning)

A **Random Forest Classifier** was trained on the customer feature store to predict whether a new buyer will become a repeat customer (is_repeat_buyer).

- **Handling Imbalance:** Applied class_weight='balanced' and stratified splitting to accurately capture the minority repeat-buyer class.
- **Feature Importance Ranking:** Extracted model feature weights, proving that purchase recency, total order count, and fulfillment performance are the strongest predictors of retention.
- **Visualization:** Generated professional Seaborn horizontal bar charts displaying relative feature importances with exact data labels.

5. Statistical Validation (Hypothesis Testing)

To back up the machine learning models with traditional inferential statistics, a **Chi-Square Test of Independence** was executed:

- H_0 **(Null Hypothesis):** Customer retention is independent of whether they experienced a delivery delay.
- H_1 **(Alternative Hypothesis):** Customer retention is dependent on delivery delays.
- **Result:** With $p < 0.05$, the null hypothesis was rejected, mathematically proving that supply chain performance has a statistically significant association with customer churn.

6. How to Run the Project

1. Clone the repository and ensure your raw CSV files are placed in the working directory.
2. Run your data ingestion script to populate ecommerce.db.
3. Execute the Pandas cleaning and Gold layer feature extraction scripts.
4. Run the Random Forest classifier and hypothesis testing scripts inside your Jupyter Notebook to reproduce metrics and visualizations.